

# ANIL PRADHAN

Full Stack Developer · AI/Agent Systems · Backend Engineering

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## EDUCATION

**B.Tech – CSE | CGPA: 9.11/10** | ITER, SOA University, Bhubaneswar 2024 – 2028  
**Class XII – PCM with IT | 85%** | Odisha Board of Secondary Education 2022 – 2024

## EXPERIENCE

**Campus Ambassador** | E-Cell, IIT Kanpur · Remote Dec 2025 – Jan 2026  
► Represented IIT Kanpur's E-Summit at ITER, SOA University; boosted student participation via tech talks and networking sessions.

## PROJECTS

**AI Career Mentor** Feb 2026 – Present  
*Next.js 14 · FastAPI · Microsoft AutoGen v0.2 · PostgreSQL · Groq · Gemini · WebSocket · Upstash Redis · Google OAuth*

- Architected a production-grade 5-agent AI platform using Microsoft AutoGen v0.2, deploying a GroupChatManager with custom speaker selection to coordinate Resume Analyst, Market Researcher, Career Coach, and LinkedIn Reviewer agents in parallel via `asyncio.to_thread()`, delivering consolidated output in under 60 seconds.
- Engineered a real-time WebSocket mock interview system using the OpenAI SDK pointed directly at the Groq API, achieving sub-2-second token streaming latency with a 7-question adaptive state machine scaling difficulty by company tier (FAANG to service-based).
- Implemented dual LLM architecture with automatic failover — Gemini 1.5 Flash as primary with programmatic Gemini-to-Groq (Llama 3.3 70B) fallback on 429 rate limit errors, ensuring zero user-facing failures.
- Built distributed rate limiting and AI response caching via Upstash Redis and SlowAPI middleware — 100 req/hr and 1000 req/day per user with SHA-256 keyed cache lookups; rate counters increment only on successful AI responses.
- Designed live market intelligence pipeline integrating DuckDuckGo tool calling inside the Market Researcher agent to fetch real-time salary ranges, top hiring companies, and in-demand skills — no static mock data.
- Deployed decoupled microservices across Vercel (Next.js 14), Render (FastAPI), Neon serverless Postgres (`pool_size=3`, `pool_recycle=300s`) with full GitHub Actions CI/CD — ESLint, pytest, and pip-audit security scanning.
- Implemented Google OAuth 2.0 end-to-end (ID Token verification via google-auth, short-lived JWT + refresh token), deterministic ATS scoring engine, and in-memory PDF parsing with pdfplumber storing structured JSON in Postgres.

**VolunteerIQ** Mar 2026 – Apr 2026  
*Next.js 14 · FastAPI · Firebase Auth · Gemini 2.5 Flash · Mapbox GL · PostgreSQL · Docker*

- Built an AI-powered NGO coordination platform for Google Solution Challenge 2026, integrating Gemini 2.5 Flash to process multi-format field surveys (PDF, CSV, DOCX, TXT) and extract urgent community problems with urgency scores and auto-generated deployable field tasks.
- Implemented AI volunteer matching system ranking candidates by skill match, location proximity, and availability with a 0–100 score, alongside a context-aware AI chat assistant querying live PostgreSQL data to answer plain-English NGO operational queries.
- Deployed a Mapbox GL live operations map with geocoding cache, Firebase JWT authentication verified server-side on every protected request, Docker + docker-compose containerization, and Gemini-to-Groq automatic fallback for uninterrupted AI service under rate limits.

## TECHNICAL SKILLS

**Languages:** Python · TypeScript · JavaScript · Java  
**Frontend:** Next.js 14 · React · Vanilla CSS · Recharts · Lucide React · Axios · react-map-gl  
**Backend:** FastAPI · Microsoft AutoGen v0.2 · OpenAI SDK (via Groq) · SlowAPI · SQLAlchemy · Alembic · Pydantic · Uvicorn · Loguru · pdfplumber · edge-tts  
**Databases:** Neon Postgres · SQLite · Upstash Redis  
**AI / LLM:** Google Gemini 1.5 Flash · Groq (Llama 3.3 70B) · Microsoft AutoGen  
**Auth & Security:** Google OAuth 2.0 · Firebase Authentication · JWT · bcrypt · google-auth  
**DevOps & Infra:** Docker · GitHub Actions · Vercel · Render.com · CI/CD  
**Tools:** VS Code · Git · Swagger UI · Postman · Mapbox GL

## ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Skills Boost Arcade Program: Earned 75+ points and multiple rewards through hands-on cloud labs and challenges focused on Google Cloud technologies.
- Major International Hackathons: Participated in Microsoft AI Dev Days Hackathon, Amazon Nova AI Challenge, and Google Solution Challenge 2026 — working continuously for multiple months on AI-based project development and problem-solving.
- Competitive Programming: Solved 200+ problems across LeetCode and GeeksforGeeks; earned a certificate and T-shirt for successfully completing the GeeksforGeeks 80 Days Coding Challenge.
- Active Tech Community: Participated in multiple offline technical workshops, coding competitions, and developer events including Google Developer Groups (GDG) sessions — improving practical knowledge in software development, AI, and modern technologies.